
Robustness of a Kernel-Based Method for the Inverse Darcy Flow Problem

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Abstract

We study an inverse Darcy flow problem in which both the permeability log-field $a(x)$ and the solution field $u(x)$ are recovered from sparse and noisy pointwise observations of u . We reproduce a simplified kernel-based method using Gaussian radial basis function expansions and a damped Gauss–Newton solver, with the PDE enforced at collocation points. On synthetic data, the method recovers the main spatial structure of both fields and converges stably, although the permeability reconstruction remains somewhat less accurate than the reference result.

1 Introduction

This project studies an inverse Darcy flow problem, where the goal is to recover both the permeability field $a(x)$ and the solution field $u(x)$ from sparse and noisy pointwise observations of u . The two unknown fields are coupled through the PDE

$$\begin{aligned} -\nabla \cdot (e^{a(x)} \nabla u(x)) &= f(x), & x \in \Omega, \\ u(x) &= 0, & x \in \partial\Omega. \end{aligned} \tag{1}$$

with $\Omega = (0, 1)^2$ and $f \equiv 1$. This setup is a standard benchmark in subsurface flow, because permeability is difficult to measure directly and has to be inferred from limited pressure or head measurements.

Such inverse problems are important in subsurface flow and porous media applications, because the permeability field is usually not directly observable and must be inferred from limited measurements. Following the kernel-based framework of Chen et al. (1), this report focuses on reproducing the core method and demonstrating that it can recover the main structures of $a(x)$ and $u(x)$ from synthetic data, and then go beyond just reproducing the results by studying how recovery degrades as a function of (i) the number of observations and (ii) the noise level.

2 Literature Review

Scientific machine learning has become an important approach for solving forward and inverse problems governed by partial differential equations (PDEs), especially when only sparse or noisy measurements are available. A broad overview of this field is given by Karniadakis et al. (3), who describe physics-informed learning as a way to combine data-driven models with prior physical knowledge. This idea is particularly useful for inverse PDE problems, where the available data may be limited but the governing equation provides important information about the admissible solution space.

One widely used method in this area is physics-informed neural networks (PINNs), introduced by Raissi et al. (4). In PINNs, the unknown solution field is represented by a neural network, and the governing PDE is enforced by adding a residual term to the loss function. The PDE residual is

usually computed through automatic differentiation, allowing the method to handle nonlinear PDEs and inverse problems without explicitly building a traditional numerical solver. This framework is attractive because it can combine sparse observations with physical constraints and can also be used to identify unknown PDE parameters.

PINN-based methods have also been applied to porous media and subsurface flow problems. Tartakovsky et al. (5) proposed a physics-informed deep neural network method for estimating hydraulic conductivity and learning constitutive relationships in saturated and unsaturated flow. Their work is closely related to our inverse Darcy problem because permeability or hydraulic conductivity is usually difficult to measure directly, while pressure or head measurements are more accessible. Their results show that adding physical constraints can improve learning when data are sparse or when direct observations of the unknown quantity are unavailable. However, their method still enforces the PDE through a soft residual penalty, so the final reconstruction depends on the optimization process and on the balance between data loss and physics loss.

Although PINNs are flexible, their training can be challenging. Wang, Yu, and Perdikaris (2) analyzed PINN training using neural tangent kernel theory and showed that fully connected PINNs can suffer from spectral bias and imbalanced convergence rates between different loss components. This means that different parts of the loss function may be learned at very different speeds, making the optimization stiff and sensitive to hyperparameters. This issue is especially relevant for inverse problems where both an unknown coefficient field and the solution field must be recovered. In our inverse Darcy setting, the solution field $u(x)$ is observed only at sparse noisy points, while the log-permeability field $a(x)$ is not directly observed and must be inferred indirectly through the PDE.

The second branch of related work is based on kernel methods and Gaussian-process (GP) collocation. Chen et al. (1) developed a unified framework for solving nonlinear PDEs and inverse problems using Gaussian processes. In their method, the unknown solution is interpreted as the maximum a posteriori estimate of a Gaussian process conditioned on satisfying the PDE at a finite set of collocation points. Through a representer theorem, the infinite-dimensional problem is reduced to a finite-dimensional optimization problem, which is then solved using a Gauss–Newton type method. Compared with PINNs, this approach has a more explicit connection to kernel collocation methods and enforces the PDE more directly at the selected collocation points.

The inverse Darcy flow experiment in Chen et al. (1) is the main reference for our project. In this problem, the goal is to recover both the solution field $u(x)$ and the log-permeability field $a(x)$ from limited noisy observations of u . Our work follows this kernel-based inverse Darcy framework but focuses on robustness. Specifically, both $u(x)$ and $a(x)$ are represented using Gaussian radial basis expansions, the Darcy PDE is enforced at interior collocation points, and the unknown coefficients are solved using a damped Gauss–Newton method. Compared with the original experiment, our contribution is to study how the reconstruction quality changes as the number of observations and the noise level vary. This is important because sparse and noisy measurements are common in real subsurface applications.

3 Methodology

We represent both the solution field $u(x)$ and the log-permeability field $a(x)$ using Gaussian radial basis expansions on a fixed center grid,

$$u(x) \approx \sum_{j=1}^M \alpha_j \phi_j(x), \quad a(x) \approx \sum_{j=1}^M \beta_j \phi_j(x), \quad (2)$$

where

$$\phi_j(x) = \exp\left(-\frac{\|x - c_j\|^2}{2\ell^2}\right). \quad (3)$$

In the baseline implementation, the length scale is set to $\ell = 0.18$, and the centers c_j are placed on a 9×9 grid inside Ω , giving $M = 81$ basis functions. This gives a finite-dimensional analogue of the Gaussian-process representation used in the reference kernel method, with the RKHS norm approximated by an ℓ_2 penalty on the expansion coefficients.

The inverse problem is then written as a nonlinear least-squares problem in the coefficient vectors (α, β) . Instead of training a neural network with soft PDE penalties, the Darcy equation is enforced

directly at collocation points, while the unknown fields are recovered jointly. The stacked residual contains four components:

1. the Darcy PDE residual at $n_\Omega = 220$ interior collocation points, using the identity

$$\nabla \cdot (e^a \nabla u) = e^a (\Delta u + \nabla a \cdot \nabla u);$$

2. the Dirichlet boundary residual $u = 0$ at $n_{\partial\Omega} = 80$ boundary points;
3. the data-misfit residual $(u(x_i) - o_i)/\gamma$ at I observation points, where

$$o_i = u^*(x_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \gamma^2);$$

4. Tikhonov-style regularization terms $\sqrt{\lambda_u}\alpha$ and $\sqrt{\lambda_a}\beta$ on the coefficients, with $\lambda_u = \lambda_a = 10^{-3}$ in the baseline case.

Equivalently, if

$$z = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

denotes the full unknown coefficient vector and $r(z)$ denotes the stacked residual, the implemented optimization problem is

$$\min_z \frac{1}{2} \|r(z)\|_2^2. \quad (4)$$

At each Gauss–Newton iteration, the Jacobian

$$J(z) = \frac{\partial r(z)}{\partial z} \quad (5)$$

is formed and the search direction is obtained from the damped normal equation

$$(J^T J + \mu I) \delta z = -J^T r. \quad (6)$$

Here μ is a Levenberg-type damping parameter used to improve numerical stability. The coefficients are updated by

$$z^{(k+1)} = z^{(k)} + \eta \delta z, \quad (7)$$

where η is chosen so that the residual loss decreases.

All derivatives appearing in the PDE residual are computed analytically from the Gaussian radial basis expansions. This avoids automatic differentiation through a neural network and preserves the main structure of the GP/kernel method in Chen et al. (1): the unknown coefficient field and the PDE solution are recovered together, and the PDE acts as a hard collocation constraint in the inverse problem. Although the implementation is simplified compared with the full paper, it preserves the main structure needed for the robustness experiments.

The reconstruction accuracy is measured using relative L^2 errors on the evaluation grid,

$$\text{Rel. } L^2(a) = \frac{\|a - a^*\|_2}{\|a^*\|_2}, \quad \text{Rel. } L^2(u) = \frac{\|u - u^*\|_2}{\|u^*\|_2}. \quad (8)$$

These metrics are used for the baseline reproduction and for all robustness and sensitivity experiments.

4 Experimental Setup and Results

4.1 Baseline Reproduction

To generate synthetic data, a ground-truth permeability field $a^*(x)$ is first prescribed, and the corresponding forward Darcy problem is solved on a fine grid to obtain the reference solution $u^*(x)$. Sparse observation points are then sampled inside the domain, and noisy measurements are generated by adding Gaussian noise to the sampled values of $u^*(x)$. This setup is consistent with the paper, where the inverse Darcy experiment also uses synthetic data generated from the forward problem rather than an external dataset. The recovered solution $u(x)$ matches the main spatial pattern of the ground truth well, while the recovered permeability field $a(x)$ captures the major high-value regions, although some local distortion remains. Figure 1 shows a single baseline result according to the prescribed methodology.

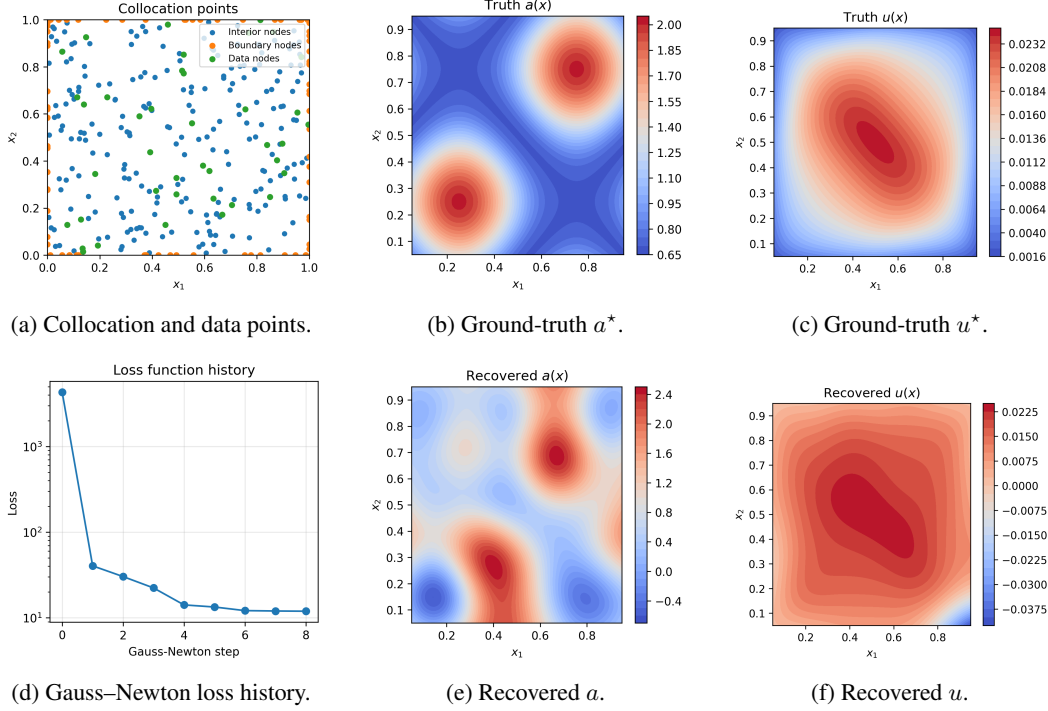


Figure 1: Baseline reproduction of the inverse Darcy experiment. The six panels show the collocation/data points, ground-truth fields, optimization history, and recovered fields using the simplified kernel-based implementation.

4.2 Observation and Noise Robustness

We now go beyond reproduction and study how the recovery behaves under varying data conditions. All experiments use the same forward solve, kernel basis, and optimizer as the baseline; only the observations and noise are changed. For each setting we run three independent trials with fresh random seeds and report mean error and one standard deviation.

Effect of the number of observations. We fix $\gamma = 10^{-3}$ and sweep the number of observation points $I \in \{5, 10, 20, 40, 80, 160\}$. The left panel of Figure 2 shows the resulting relative L^2 errors. The error in u decreases steadily as more observations are added, falling from roughly order-one error at $I = 5$ to approximately 3×10^{-2} at $I = 160$. This behavior is expected because u is the directly observed field, so increasing the number of data points directly improves its reconstruction. The error in a also decreases, but more slowly. It drops from roughly 1.1–1.2 at very small data budgets to about 0.35–0.45 for the larger observation counts. Unlike u , the permeability reconstruction shows diminishing returns once a moderate number of observations is available. This indicates that the remaining error in a is not controlled only by data availability, but also by the indirect nature of the inverse problem and by the kernel/regularization choices.

Effect of observation noise. We then fix $I = 40$ and sweep the noise standard deviation $\gamma \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. The right panel of Figure 2 shows the result. For $\gamma \leq 10^{-3}$, the reconstruction remains relatively stable: the error in u stays near 10^{-1} , while the error in a remains in the 0.45–0.55 range. When the noise level is increased to $\gamma = 10^{-2}$, both errors increase sharply, with the error in u rising to roughly 0.7 and the error in a reaching about 1.0. At $\gamma = 10^{-1}$, the reconstruction degrades substantially, especially for u , whose relative error grows to order 2. This shows that the method is robust to small observational noise but breaks down once the noise becomes comparable to, or larger than, the scale of the observed solution values.

Recovery of u behaves like a standard regression problem: it improves steadily with more data and with less noise. Recovery of a does not: beyond a small data budget it is limited by prior/regularization choices and it is the first thing to break once noise becomes comparable to the magnitude of u itself. This is consistent with the role of strong priors in the inverse Darcy recovery problem in the reference

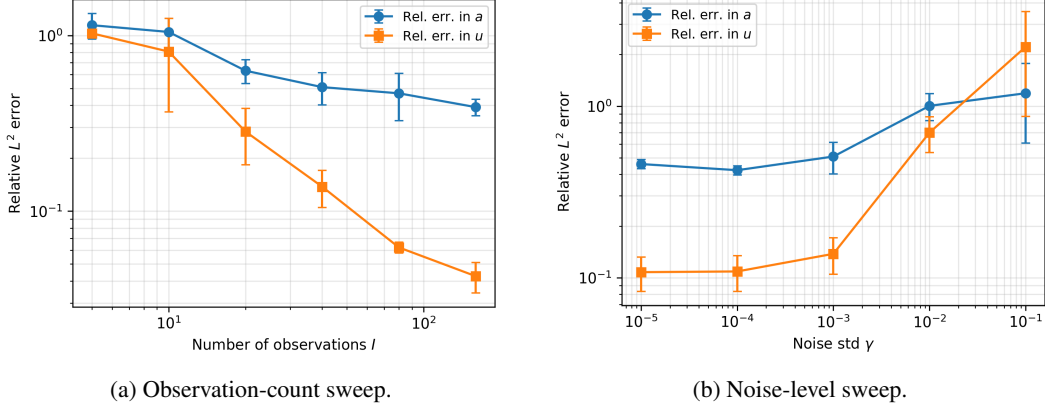


Figure 2: Robustness sweeps. Left: relative L^2 error in a and u versus number of observation points I at fixed noise $\gamma = 10^{-3}$. Right: relative error versus observation noise standard deviation γ at fixed $I = 40$. Error bars are one standard deviation over three random trials.

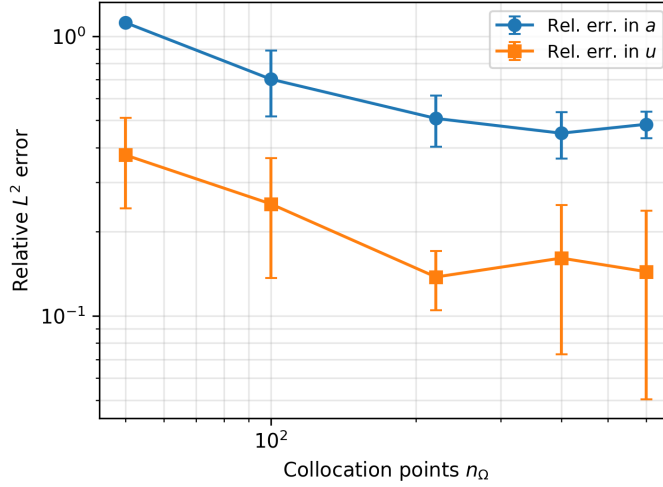


Figure 3: Effect of the number of interior collocation points n_Ω on the relative L^2 reconstruction error. Error bars represent one standard deviation over three random trials.

kernel framework (1), and it is the kind of degradation that is difficult to diagnose cleanly in soft-constraint methods such as PINNs in comparable regimes (2).

4.3 Collocation and Parameter Sensitivity

In addition to the observation-number and noise-level sweeps, we further tested how the numerical setup affects the inverse reconstruction. The next experiment varies the number of interior collocation points n_Ω . These are the points where the Darcy PDE residual is enforced. Using more collocation points gives the optimizer more information about the PDE constraint, although it also increases the size of the nonlinear least-squares problem.

Figure 3 shows that increasing the number of collocation points generally reduces the reconstruction error. The improvement is most obvious when n_Ω is small, because the PDE constraint is not sufficiently enforced with too few collocation points. As n_Ω increases, both the error in $a(x)$ and the error in $u(x)$ decrease. The error in $a(x)$ drops from above 1 to around 0.45–0.5, while the error in $u(x)$ decreases to around 0.13–0.16. However, the improvement becomes weaker when n_Ω is already large, and the curves show small fluctuations at the largest tested values. This suggests a diminishing-return effect: after the PDE has been sufficiently sampled, adding more collocation

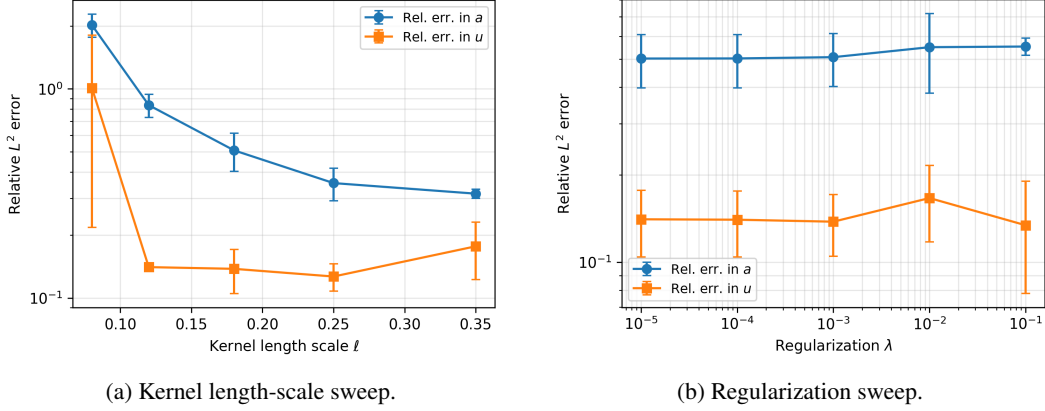


Figure 4: Parameter sensitivity studies. Left: relative L^2 error versus Gaussian kernel length scale ℓ . Right: relative L^2 error versus regularization strength λ , with $\lambda_u = \lambda_a = \lambda$. Error bars represent one standard deviation over three random trials.

points does not strongly improve the reconstruction. The error in $a(x)$ remains higher than the error in $u(x)$, which again shows that the log-permeability field is harder to recover because it is not directly observed.

The last two experiments study the sensitivity of the method to two model parameters: the Gaussian kernel length scale ℓ and the regularization strength λ . In the length-scale sweep, only the width of the Gaussian radial basis functions is varied. In the regularization sweep, the coefficient penalties are varied together by setting $\lambda_u = \lambda_a = \lambda$.

The length-scale sweep in Figure 4 shows that the reconstruction is sensitive to the choice of ℓ . When ℓ is too small, the Gaussian basis functions are too localized, and the reconstruction becomes unstable. As ℓ increases, the errors in both $a(x)$ and $u(x)$ decrease in this experiment, which indicates that a smoother basis is more suitable for the prescribed synthetic permeability field. However, for more complex permeability fields, an excessively large length scale could oversmooth the spatial variation, so the best choice of ℓ should depend on the expected scale of the true coefficient field.

The regularization sweep in Figure 4 shows a weaker dependence on λ . Over the tested range from $\lambda = 10^{-5}$ to $\lambda = 10^{-1}$, the error curves remain relatively flat. The error in $a(x)$ stays roughly in the 0.5–0.6 range, and the error in $u(x)$ stays around 0.12–0.17. This suggests that, in the current simplified setup, the Gaussian basis representation and PDE collocation constraint already provide substantial stabilization. The Tikhonov regularization mainly prevents unstable coefficient growth, but it does not dominate the reconstruction accuracy in this experiment.

Overall, these additional experiments show that the reconstruction is more sensitive to the number of PDE collocation points and the kernel length scale than to the regularization strength. Across all experiments, $u(x)$ is recovered more accurately than $a(x)$, which is consistent with the structure of the inverse problem: $u(x)$ is directly observed, while $a(x)$ must be inferred indirectly through the PDE.

5 Conclusion

In this project, we studied a simplified kernel-based method for the inverse Darcy flow problem. The goal was to recover both the solution field $u(x)$ and the log-permeability field $a(x)$ from limited noisy observations of u , while using the PDE to guide the reconstruction.

The results show that $u(x)$ was recovered more accurately than $a(x)$, which is expected since the observations were taken directly from u and a had to be inferred indirectly. Increasing the number of observation points generally improved the reconstruction, while large observation noise made the results worse. The collocation point study showed that using too few points weakens the PDE constraint, although adding more points did not always lead to improvement.

The parameter studies showed that the kernel length scale had a noticeable effect on the reconstruction, while changing the regularization parameter over the tested range had very little effect. Overall, the simplified method was able to recover the solution field reasonably well, but the permeability field remained harder to identify. Future work could use more trials, more systematic parameter tuning, and a closer implementation of the full Gaussian-process method from the reference paper.

Author Contributions

The work for this project was divided equally between the two team members. Xu developed the baseline implementation and produced the baseline numerical results used as the starting point for the robustness study. Harshavardhan extended this baseline code to perform the four proposed robustness experiments, including the studies on the number of observation points, observation noise, collocation-point variation, and kernel and regularization sensitivity study.

For the written report, Harshavardhan wrote the abstract, introduction, methodology section, conclusion, and the discussion for the first two experimental studies. Xu wrote the Literature Review section and the discussion for the remaining experimental studies.

AI Usage Statement

AI assistance was used for language editing, generating and formatting plotting code, and debugging Python code for the experiments. Specifically, ChatGPT was used to help write Python code to visualize the inverse Darcy reconstruction results using Matplotlib, adjust plot formatting such as figure size, axis labels, legends, and layout, and debug coding errors during implementation. The mathematical formulation, numerical experiments, interpretation of results, and final submitted report were reviewed and completed by the authors.

Example prompts used include: “For this inverse Darcy method, for this Implementation [CODE PASTED] can you help me write Python code to generate the result figures?”, “Can you help me adjust the formatting of this plotting code so that the axis labels and legends are clearer?”, and “Here is the code: [CODE]. Can you figure out the issue or error in this code, debug it, and explain where we went wrong?”

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